

### 第三章习题参考答案

3.1 以下几个量的量纲是什么？

a)  $\mathbf{E} \cdot \mathbf{D}$   $\text{J/m}^3$ ; b)  $\mathbf{H} \cdot \mathbf{B}$   $\text{J/m}^3$ ; c)  $\mathbf{S}$   $\text{W/m}^2$

3.2

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} = 0, \because \mathbf{J} = 0, \therefore \frac{\partial \mathbf{D}}{\partial t} = 0, \text{ 所以 } \mathbf{D} \text{ 随时间不变化。}$$

3.3

$$\begin{aligned} \nabla \cdot \mathbf{D}_t &= \rho_{vt} = \rho_{v1} + \rho_{v2} = \nabla \cdot \mathbf{D}_1 + \nabla \cdot \mathbf{D}_2 = \nabla \cdot (\mathbf{D}_1 + \mathbf{D}_2) \\ \nabla \times \mathbf{H}_t &= \mathbf{J}_t + \frac{\partial \mathbf{D}_t}{\partial t} = (\mathbf{J}_1 + \mathbf{J}_2) + \frac{\partial (\mathbf{D}_1 + \mathbf{D}_2)}{\partial t} \\ &= \mathbf{J}_1 + \frac{\partial \mathbf{D}_1}{\partial t} + \mathbf{J}_2 + \frac{\partial \mathbf{D}_2}{\partial t} = \nabla \times \mathbf{H}_1 + \nabla \times \mathbf{H}_2 = \nabla \times (\mathbf{H}_1 + \mathbf{H}_2) \end{aligned}$$

$$\mathbf{B} = \mu \mathbf{H} \quad \mathbf{D} = \varepsilon \mathbf{E}$$

所以，麦克斯韦方程的解为  $(\mathbf{E}_1 + \mathbf{E}_2, \mathbf{B}_1 + \mathbf{B}_2, \mathbf{H}_1 + \mathbf{H}_2, \mathbf{D}_1 + \mathbf{D}_2)$

3.4

由斯托克斯定理  $\oint_l \mathbf{E} \cdot d\mathbf{l} = \int_s (\nabla \times \mathbf{E}) \cdot d\mathbf{S}$

$$\mathbf{E} = 0 \quad \int_s (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

所以  $\int_s \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} = 0$ , 不能得出  $\frac{\partial \mathbf{B}}{\partial t} = 0$

3.5

$$\omega_p = \sqrt{\frac{Ne^2}{m\varepsilon_0}} = 5.64 \times 10^7; \quad \frac{\varepsilon_e}{\varepsilon_0} = 1 - \frac{\omega_p^2}{\omega^2}$$

$$\text{当 } \omega = 0.5 \text{ MHz}, \quad \frac{\varepsilon_e}{\varepsilon_0} = 1 - \frac{31.7 \times 10^{14}}{(2\pi \times 0.5 \times 10^6)^2} = -320.5$$

$$\text{当 } \omega = 1 \text{ MHz}, \quad \frac{\varepsilon_e}{\varepsilon_0} = 1 - \frac{31.7 \times 10^{14}}{(2\pi \times 10^6)^2} = -79.1$$

所以电离层有效介电系数  $\varepsilon_e$  的变化范围为  $-320.5\varepsilon_0 < \varepsilon_e < -79.1\varepsilon_0$

## 3.7

为了计算方便，设  $t = 0$  时  $\varphi = 0$ ，而  $\varphi = \omega t$ ，点电荷  $q$  在  $O$  点产生电位移矢量

$D$  为：

$$\mathbf{D} = \frac{q}{4\pi r^2}(-\mathbf{r}_0) = \frac{q}{4\pi r^2}(-\mathbf{x}_0 \cos \varphi - \mathbf{y}_0 \sin \varphi) = \frac{q}{4\pi r^2}(-\mathbf{x}_0 \cos \omega t - \mathbf{y}_0 \sin \omega t)$$

位移电流密度为

$$\mathbf{J}_{dx} = \frac{\partial D_x}{\partial t} = \mathbf{x}_0 \frac{q\omega}{4\pi r^2} \sin \omega t$$

$$\mathbf{J}_{dy} = -\mathbf{y}_0 \frac{q\omega}{4\pi r^2} \cos \omega t$$

把数值代入上式：

$$\mathbf{J}_d = \frac{25}{\pi}(\mathbf{x}_0 \sin 1000t - \mathbf{y}_0 \cos 1000t)$$

## 3.9

$$\mathbf{E}(t) = \cos(\omega t - z)\mathbf{x}_0 - \sin(\omega t - z)\mathbf{y}_0$$

$$\mathbf{H}(t) = \cos(\omega t - z)\mathbf{x}_0 + \sin(\omega t - z)\mathbf{y}_0$$

$$\mathbf{S}(t) = \mathbf{E}(t) \times \mathbf{H}(t) = \begin{vmatrix} \mathbf{x}_0 & \mathbf{y}_0 & \mathbf{z}_0 \\ \cos(\omega t - z) & -\sin(\omega t - z) & 0 \\ \sin(\omega t - z) & \cos(\omega t - z) & 0 \end{vmatrix} = \mathbf{z}_0$$

$$\langle \mathbf{S}(t) \rangle = \mathbf{z}_0$$

## 3.10

$$\mathbf{H} = -\frac{1}{j\omega\mu_0} \nabla \times \mathbf{E} = -\frac{\mathbf{z}_0}{j\omega\mu_0} \frac{m\pi E_{ym}}{a} \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$$

$$\begin{aligned} \mathbf{S}(t) &= \mathbf{E}(t) \times \mathbf{H}(t) = \mathbf{y}_0 E_{ym} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} \cos \omega t \times \frac{-\mathbf{z}_0}{\omega\mu_0} \frac{m\pi E_{ym}}{a} \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b} \sin \omega t \\ &= -\mathbf{x}_0 \frac{m\pi E_{ym}^2}{4a\omega\mu_0} \sin \frac{2m\pi x}{a} \cos^2 \frac{n\pi y}{b} \sin 2\omega t \end{aligned}$$

$$\langle \mathbf{S}(t) \rangle = \frac{1}{2} \operatorname{Re}(\mathbf{E} \times \mathbf{H}^*) = \frac{1}{2} \operatorname{Re} \left( \mathbf{y}_0 E_{ym} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b} \times \frac{-\mathbf{z}_0}{j\omega\mu_0} \frac{m\pi E_{ym}}{a} \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{b} \right) = 0$$